

Hardware Wiring

AquaSensePhase1—CompleteCircuitWiringandPin Mapping

This document defines the finalized wiring for the Arduino Uno, I2C LCD, HC-SR04 ultrasonic sensor, two servo motors, and the 18650 battery power system.

Finalized Pin Mapping

Module	HC-	Connection	Arduino Uno pin / rail
SR04	HC-	TRIG	D9
SR04	HC-	ECHO	D10
SR04	HC-	VCC	5V
SR04 Servo 1		GND	GND
Servo 2		Signal / PWM	D6
I2C LCD 16x2		Signal / PWM	D7
I2C LCD 16x2		SDA	A4 / SDA
I2C LCD 16x2		SCL	A5 / SCL
I2C LCD 16x2		VCC	5V or regulated 5V
		GND	GND

1. Arduino Uno to I2C LCD Display

- LCD GND → Arduino GND.
- LCD VCC → Arduino 5V or a suitable regulated 5V supply.
- LCD SDA → Arduino A4 / SDA.
- LCD SCL → Arduino A5 / SCL.
- The LCD address used by the program is 0x27. If the display does not respond, verify the module address.

2. Arduino Uno to HC-SR04

- HC-SR04 VCC → Arduino 5V.
- HC-SR04 GND → Arduino GND.
- HC-SR04 TRIG → Arduino D9.
- HC-SR04 ECHO → Arduino D10.
- The program uses a hand-detection threshold of 15 cm.

3. Arduino Uno to Servo Motors

- Servo 1 signal / PWM ↔ Arduino D6.
- Servo 2 signal / PWM → Arduino D7.
- Servo power VCC → suitable regulated servo supply derived from the battery system.
 - Servo grounds → power-supply negative and common Arduino GND.

- Both servos are commanded to 90° for the open position and 0° for the closed position in the Phase 1 program.

4. Power Distribution

- 18650 battery-pack positive (+)
→ protected and regulated power path for the system.
- 18650 battery-pack negative (-) → system common ground.
- The Arduino, sensor, LCD, and servo supply must share a common reference ground.
- If the battery voltage is applied to Arduino VIN, ensure that the battery-pack voltage and regulator arrangement are appropriate for VIN. Do not connect an unregulated battery voltage directly to the 5V pin.
- The servo supply must be capable of handling the combined peak current of both SG90 servos.

5. Common Ground

A common ground is required for reliable signal operation. Connect Arduino GND, LCD GND, HC-SR04 GND, servo-supply negative, and battery negative to the same ground reference. The servo power may be supplied separately, but its ground must remain connected to Arduino GND.

6. Wiring Summary

Signal / power line	Final connection
LCD SDA	A4 / SDA
LCD SCL	A5 / SCL
HC-SR04 TRIG	D9
HC-SR04 ECHO	D10
Servo 1 signal	D6
Servo 2 signal	D7
Sensor and LCD VCC	5V regulated rail
Servo VCC	Separate suitable regulated servo rail
All grounds	Common GND

7. Safety and Testing Notes

- Disconnect battery power before changing wiring.
- Use a protected 18650 battery pack, suitable battery holder, and compatible charger.
- Keep water away from the battery, regulator, Arduino, LCD backpack, and exposed connectors.
- Test the circuit first without water and verify servo direction, valve travel, LCD operation, and sensor readings.
- Check polarity and voltage with a multimeter before connecting the battery.